

Phetolo Tshukudu

Brixton, Johannesburg, South Africa | 071 800 6606 | phetolotshukudu@gmail.com

LinkedIn: [linkedin.com/in/phetolo-tshukudu-96a8a8249](https://www.linkedin.com/in/phetolo-tshukudu-96a8a8249) | GitHub: github.com/phetolo-tshukudu

PROFESSIONAL SUMMARY

Final-year BSc Mathematical Sciences student at the University of Johannesburg, majoring in Computer Science and Mathematical Statistics, with a strong foundation in software engineering, backend development and data analysis. Hands-on experience building Java/Spring Boot REST APIs and Python data-analysis applications, with practical experience in relational databases, object-oriented programming, automated testing and statistical analysis. Interested in graduate and junior software engineering, backend development and data engineering opportunities where I can apply strong programming, analytical and problem-solving skills to build reliable, data-driven systems.

EDUCATION

University of Johannesburg

2023 - Present

BSc. Mathematical Sciences in Computer Science and Mathematical Statistics

Relevant coursework

Object-Oriented Programming • Data Structures & Algorithms • Database Systems • Software Engineering • Computer Networks • Operating Systems • Mathematical Statistics • Statistical Modelling • Regression Analysis

Schooling /Career Highlights

-
- High school learner council
 - Academic award in Mathematics
 - UJ UJenius Certificate

SKILLS

-
- **Languages:** Java, Python ,SQL, R
 - **Backend:** Spring Boot, Spring Security, REST APIs, JWT, JPA/Hibernate
 - **Python & Data:** Pandas, Data Cleaning, Data Analysis, Statistical Analysis, Dataclasses
 - **Databases:** MySQL, SQLite, Relational Database Design
 - **Testing:** Pytest, JUnit 5, Mockito
 - **Development Tools:** Git, GitHub, Maven, Eclipse IDE, Postman, MySQL Workbench
 - **Core Computer Science:** Data Structures & Algorithms, Graph Algorithms, Database Systems, Software Engineering
 - **Statistics & Data:** Statistical Modelling, Regression Analysis, Data Analysis, Data Interpretation & Visualisation, Model Evaluation & Validation

WORK EXPERIENCE (PART TIME WORK)

MATHEMATICS & PHYSICAL SCIENCES TUTOR Ghetto Hero Saturday School | Jun 2023 – Jul 2023

- Tutored Grade 10–12 learners in Mathematics and Physical Sciences.

- Explained complex concepts using structured, accessible problem-solving approaches.
- Supported learners in developing mathematical reasoning and problem-solving skills.

LEADERSHIP EXPERIENCE

TEAM LEADER — FREE SANITARY PAD & TOILETRY PROJECT Bridge to Employment Programme, NICDAM

- Led a team in organising a community project focused on providing sanitary products and toiletries.
- Coordinated team responsibilities and project activities to support successful project delivery.
- Engaged potential sponsors to seek donations and support for the initiative.
- Developed practical experience in leadership, teamwork, communication and project coordination.

FINCORE - Production-Ready Personal Finance Backend

Java 21 | Spring Boot 3.5 | Spring Security | JWT | PostgreSQL | Flyway | Docker | OpenAPI | Maven

- Built a secure REST API for managing users, accounts, transactions and personal budgets.
- Implemented JWT authentication, PostgreSQL persistence, Flyway migrations, validation, API documentation and Docker support.

FINANCIAL DATA ANALYZER

Python | Pandas | Pytest | Dataclasses | Statistics | Git/GitHub

- Built a Python application for cleaning, processing and analysing financial transaction data.
- Added spending statistics, budget tracking and 16 automated pytest tests for core business logic.

PERSONALFINANCE API - Secure Financial Management Backend

Java | Spring Boot | Spring Security | JWT | MySQL | JPA/Hibernate | JUnit 5 | Mockito | Maven

- Developed a REST API for users, accounts, transactions and budgets using a layered Spring Boot architecture.
- Implemented JWT security, JPA/Hibernate persistence, validation, exception handling and unit tests.

WILDFIRE SIMULATOR

Java | JavaFX | Data Structures & Algorithms | Graph Algorithms | Dijkstra's Algorithm | Git

- Collaborated on a Java application that simulates wildfire spread and identifies safer escape routes.
- Used Dijkstra's algorithm, priority queues and custom data structures to model routes and fire conditions.